

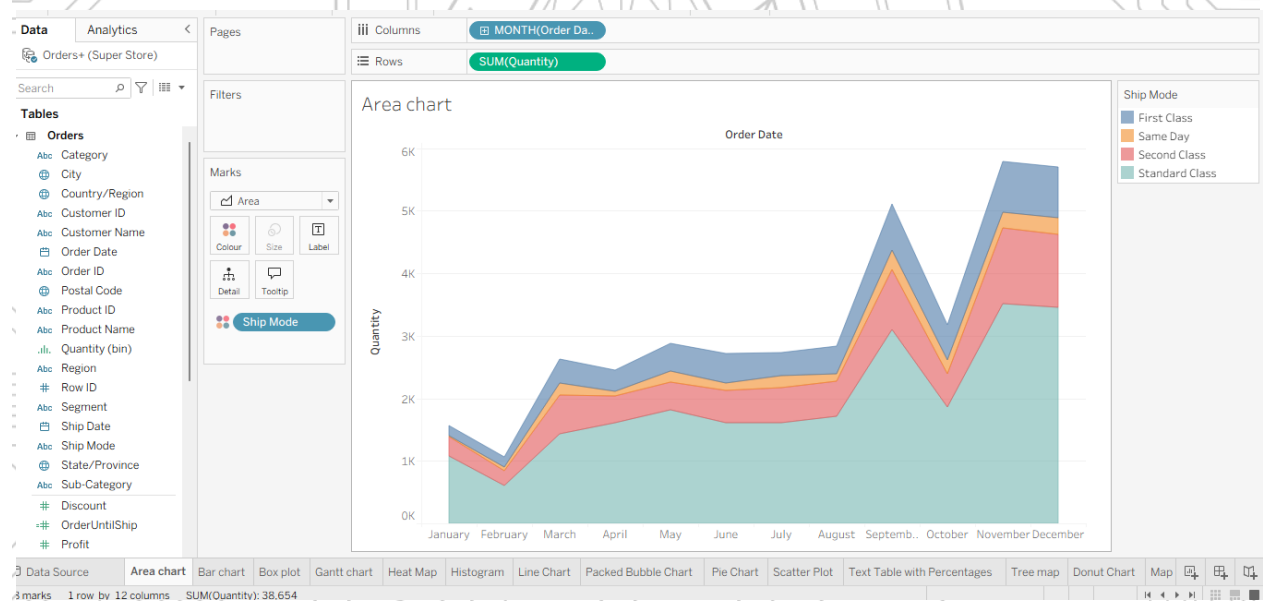
ASSIGNMENT

HALL TICKET NO	0225046019
NAME OF THE STUDENT	KAMASANI PREMCHANDU
LEARNING TRACK	DATA ANALATICS WITH TABLEAU

Sheet 1: Area chart :

1. Drag Order Date to Columns, then set it to Month.
2. Drag Quantity to Rows.
3. Drag Ship Mode to Color.
4. On the Marks card, change Automatic to Area.

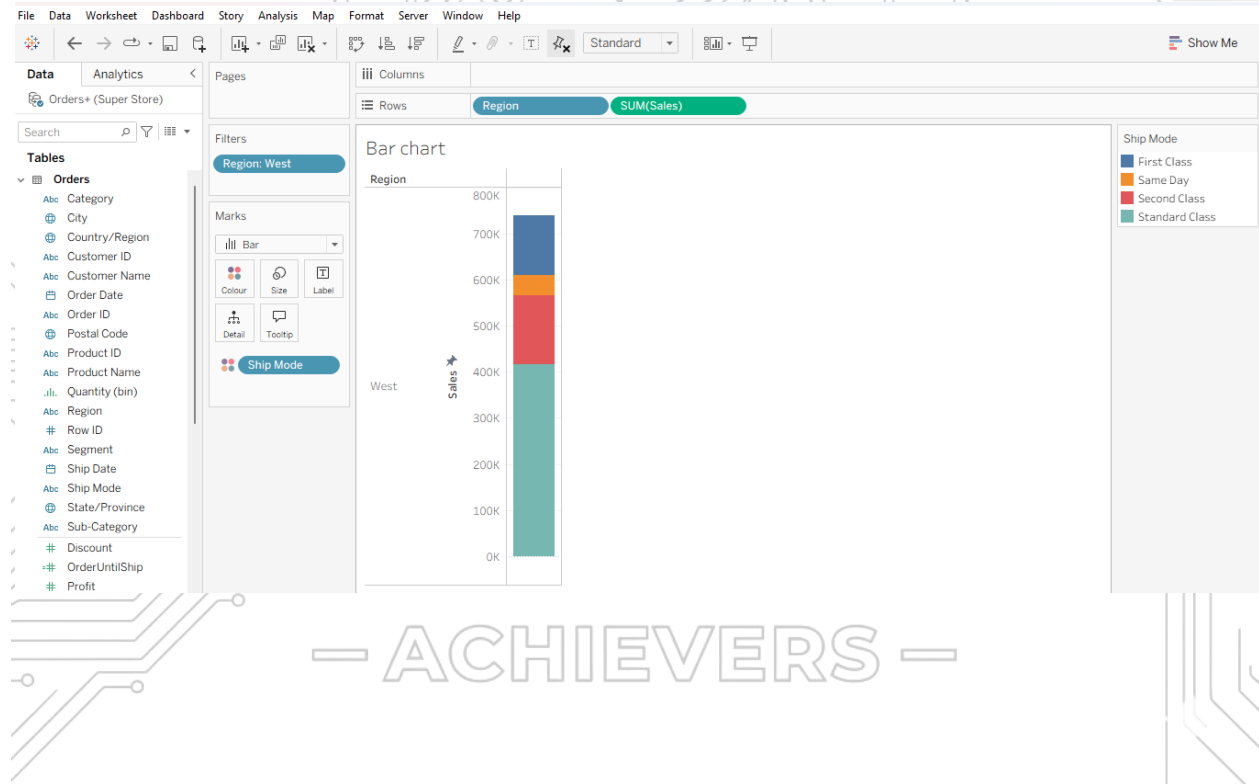
Out put:



Sheet 2: bar chart

1. Add Sales: Drag Sales to Rows. It becomes SUM(Sales) and shows as a line by year.
2. Change to Bars: On the Marks card, change Automatic to Bar.
3. Color by Ship Mode: Drag Ship Mode to Color on the Marks card. Each bar gets split by ship mode
4. Add Region: Drag Region to Rows, to the left of SUM(Sales). Now you see bars for Central, East, South, West.
5. Filter to West: Drag Region to Filters. Uncheck Central, East, South. Leave only West checked. Click OK.

Out put:



Sheet 03 : Box Plot

1. Set up the basic view

- Drag Segment to Columns

- Drag Discount to Rows

2. Change Discount to Average

- Click Discount on Rows

- Change aggregation from `SUM` to `AVERAGE`

3. Add Region breakdown

- Drag Region to Columns and place it to the right of Segment

4. Turn it into a Box Plot

- Open Show Me in the toolbar

- Select Box-and-Whisker Plot

5. Fix Region placement

- If Region moved to the Marks card, drag it back to *Columns* to the right of Segment

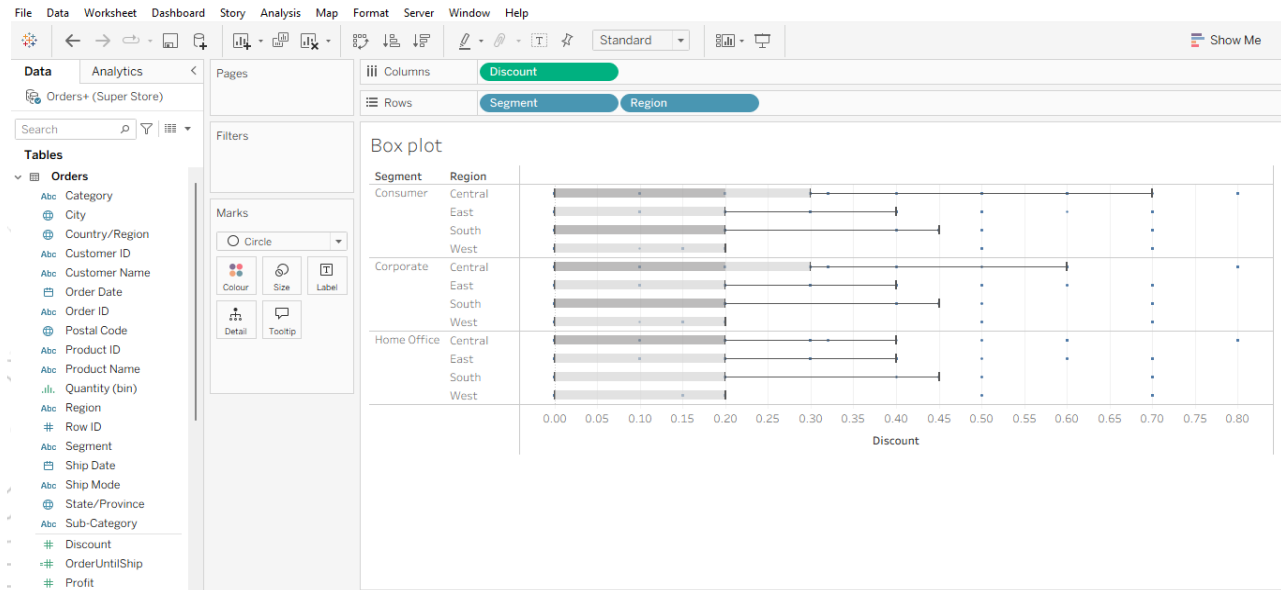
6. Show individual data points

- Go to Analysis > uncheck Aggregate Measures

7. Flip to horizontal view

- Click the Swap Rows and Columns button

Out put:



Sheet 4: Gantt chart

1. Set the timeline

- Drag Order Date to Columns. Tableau defaults to `YEAR(Order Date)`.
- Click the dropdown on `YEAR(Order Date)` and change it to Week Number. Now you have a weekly timeline.

2. Add what you're tracking

- Drag Sub-Category to Rows.
- Drag Ship Mode to Rows and place it to the right of Sub-Category.

3. Change to Gantt view

- On the Marks card, change `Automatic` to Gantt Bar. You'll see bars stretching across the weeks.

4. Create the delay calculation

- Go to Analysis > Create Calculated Field.
- Name it `OrderUntilShip`.

- Formula: `DATEDIFF('day', [Order Date], [Ship Date])`

- Click OK.

5. Make bars show delay length

- Drag `OrderUntilShip` to Size on the Marks card. Longer bars = longer shipping delay.

6. Show average delay

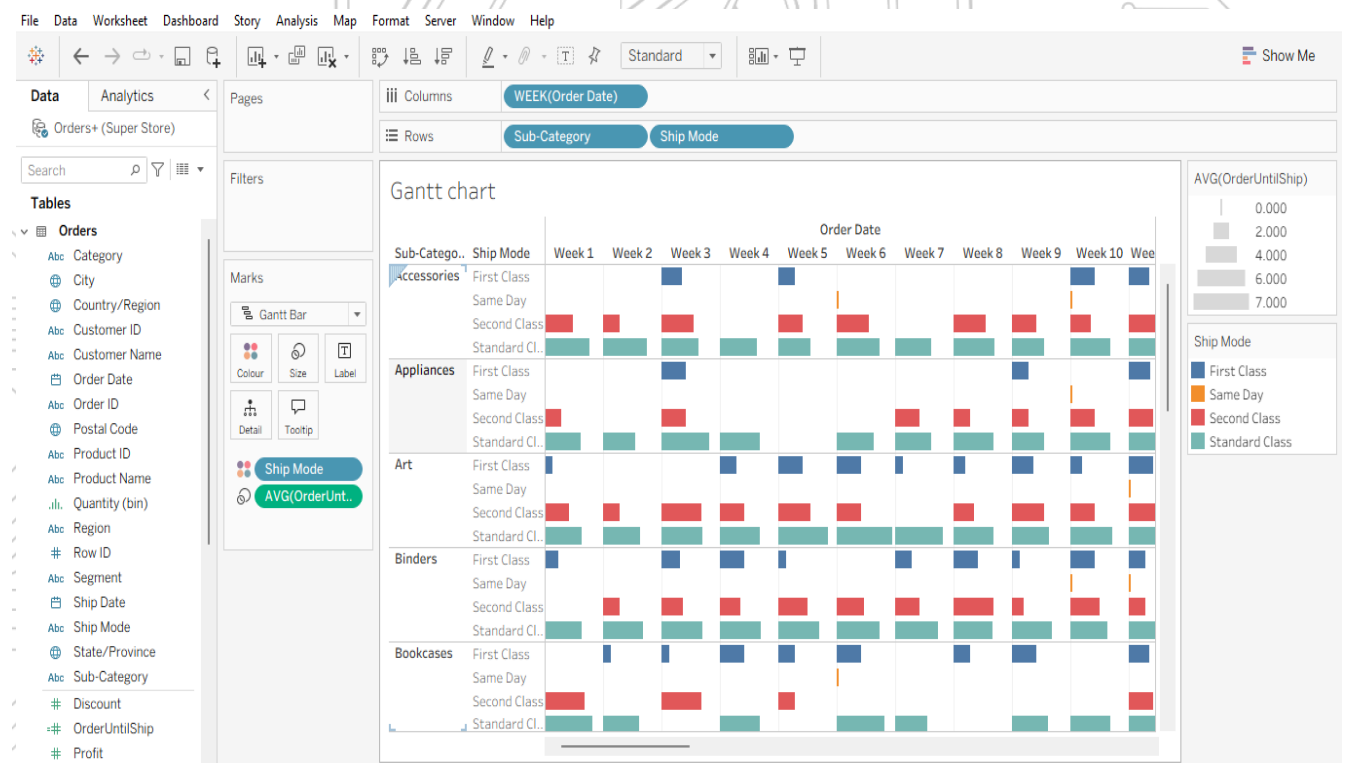
- Right-click `SUM(OrderUntilShip)` on the Marks card.

- Change aggregation to Average. Now the size reflects average shipping delay.

7. Color by Ship Mode

- Drag Ship Mode to Color on the Marks card. Each ship mode gets its own color.

Out put:



Sheet 05 : Heat Map

1. Set up the grid

- Drag Segment to Columns. You'll get Consumer, Corporate, Home Office across the top.
- Drag Region to Rows.
- Drag Sub-Category to Rows and place it to the right of Region. Now each region breaks down into sub-categories.

2. Color by Profit

- Drag *Profit* to *Color* on the Marks card.
- By default: higher profit = darker blue, losses = orange.

3. Make cells easier to read

- Click *Color* on the Marks card.
- Under Border, choose Medium Gray. This adds borders around each box.

4. Change the color scheme

- Click the dropdown arrow on the *Profit* color legend > Edit Colors.
- Set *Palette* to `Red-Black Diverging`.
- Check Use Full Color Range > Apply > OK.
- Now: profit = black/dark, loss = red.

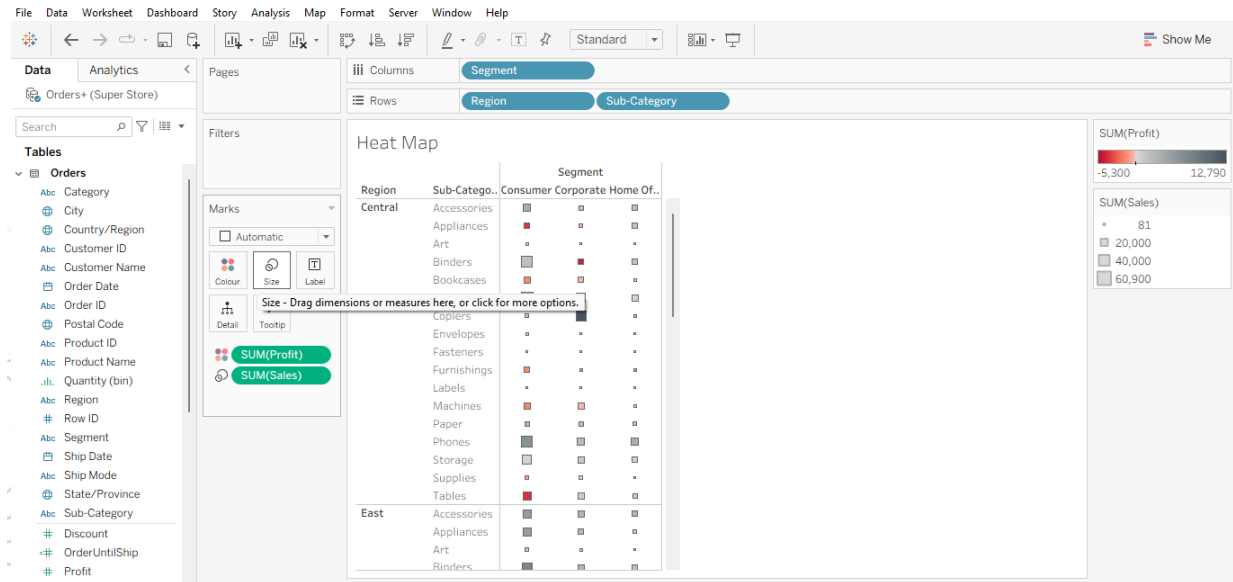
5. Show sales with box size

- Drag Sales to Size on the Marks card.
- Bigger box = higher sales, smaller box = lower sales.

6. Adjust box size

- Click Size on the Marks card.
- Move the slider right until the boxes fit and look clean.

Out put:



Sheet 06 : Histogram

1. Start the histogram

- Drag Quantity to Columns.
- Open Show Me in the top right and select Histogram.
- Tableau auto-creates 'Quantity (bin)' on Columns, moves 'CNT(Quantity)' to Rows, and sets the chart type.

2. Understand what you're seeing

- X-axis: Bins of quantity values like 1, 2, 3, etc.
- Y-axis: Number of orders in each bin.
- Example: A tall bar at quantity 2 means a lot of orders had 2 items.

3. Break it down by Segment

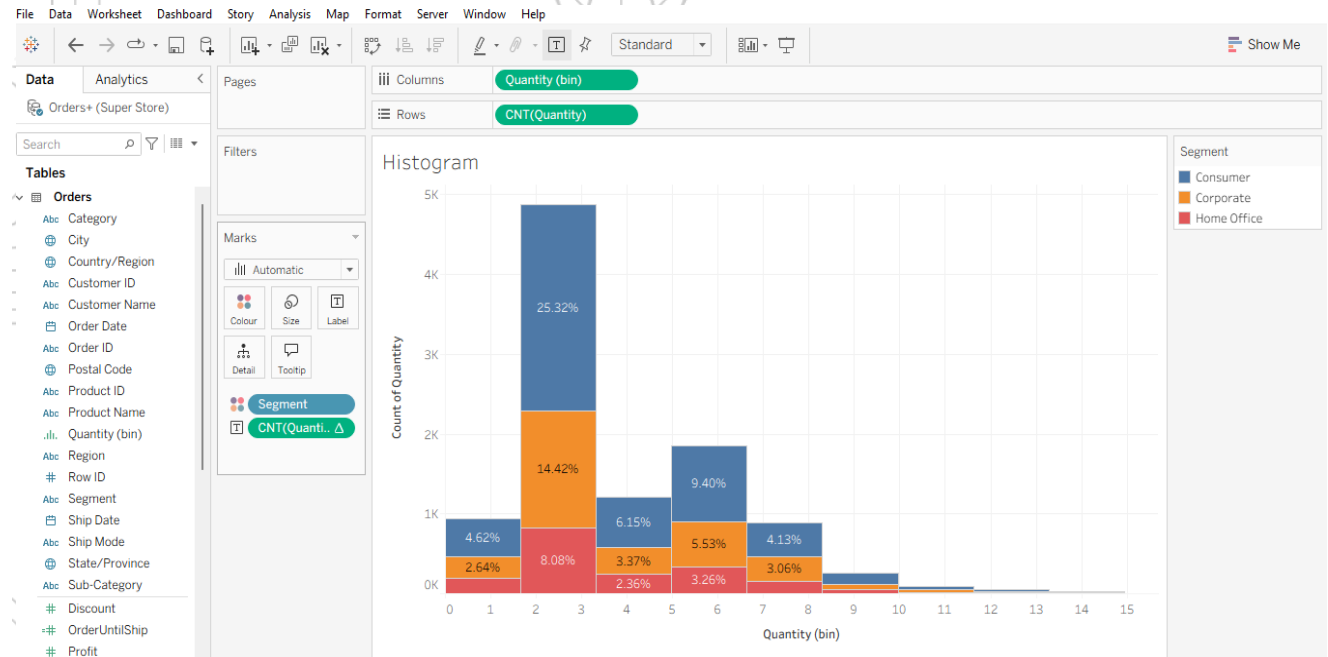
- Drag Segment to Color on the Marks card.
- Each bar now stacks to show Consumer, Corporate, and Home Office contributions.

4. Add counts or percentages to bars

- Hold Ctrl and drag `CNT(Quantity)` from Rows to Label on the Marks card. This copies the field and shows counts on bars.

- To show percentages instead: right-click `CNT(Quantity)` on the Marks card > Quick Table Calculation > Percent of Total.

Out put:



Sheet 07 : Line Chart

1. Set the time axis

- Drag Ship Date to Columns. Tableau defaults to `YEAR(Ship Date)`.

2. Plot Sales and Profit

- Drag Sales to Rows → you get a line chart for sales by year.

- Drag Profit to Rows, placing it to the right of Sales. Now you have two separate axes.

3. Combine into one axis

- Drag `SUM(Profit)` from Rows and drop it directly onto the Sales axis.

- Both lines now share the same scale, making them easy to compare.

4. Switch to monthly view

- Click the dropdown on `YEAR(Ship Date)` in Columns.

- Change it to Month from the continuous section.

- Now you see monthly trends across all years.

5. Read the trend

- Look for patterns like higher values at year-end and repeating yearly spikes.

- This is called seasonality – regular sales/profit cycles tied to time of year.

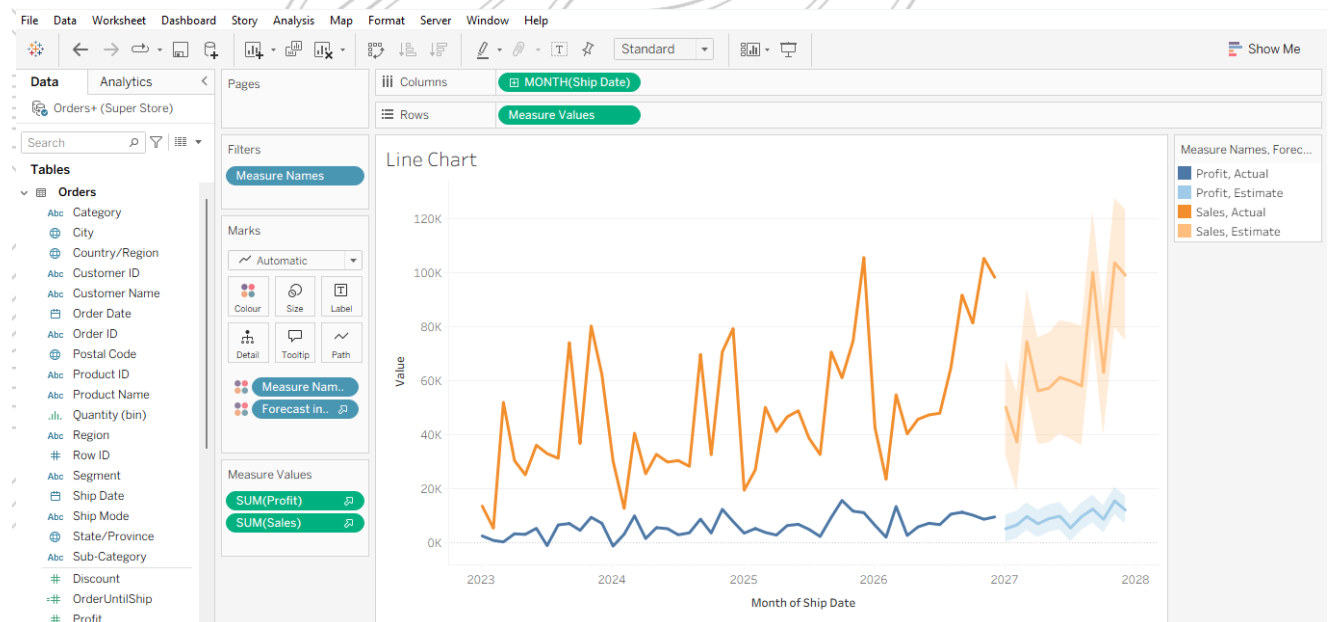
6. Add a forecast

- Open the Analytics pane on the left.

- Drag Forecast into the chart area.

- Tableau will add a predicted trend line based on your data.

Out put:



Sheet 08 : Packed Bubble Chart

1. Set up the base

- Drag Category to Columns. You'll see Furniture, Office Supplies, Technology.
- Drag Sales to Rows. Tableau makes a bar chart by default.

2. Switch to bubbles

- Open Show Me in the top right and select Packed Bubbles.
- Now each Category becomes a bubble, and bubble size = total Sales.

3. Break bubbles down by Region

- Drag Region to Detail on the Marks card.
- Each Category bubble splits into smaller bubbles for East, West, Central, and South.

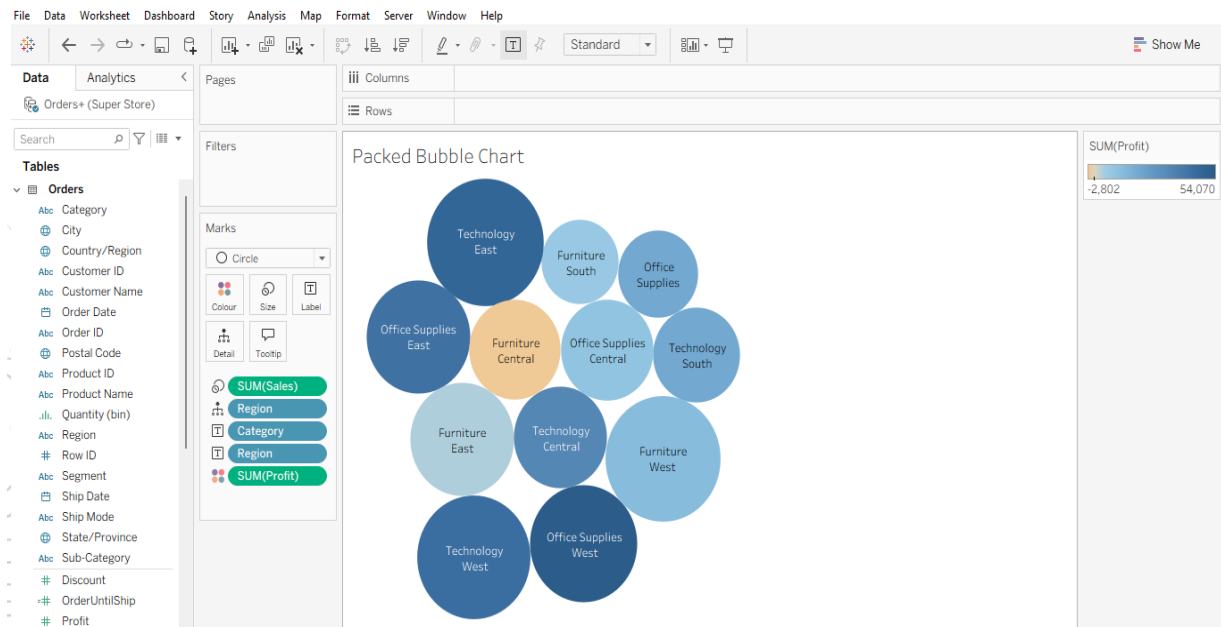
4. Color by Profit

- Drag Profit to Color on the Marks card.
- Bubble color now shows profit level:
 - Dark blue = high profit
 - Light blue = lower profit
 - Orange/Red = loss, if you change the palette

5. Label the bubbles

- Drag Region to Label on the Marks card.
- Each small bubble will now display its region name.

Out put:



Sheet 09 : Pie Chart

1. Set up the base chart

- Drag Profit to Columns.
- Drag Category to Rows. Tableau defaults to a bar chart.

2. Switch to pie chart

- Open Show Me in the top right and select Pie Chart.
- Tableau automatically updates:
 - Mark type changes to Pie
 - SUM(Profit) goes to Angle and Size
 - Category goes to Color

3. Make it bigger

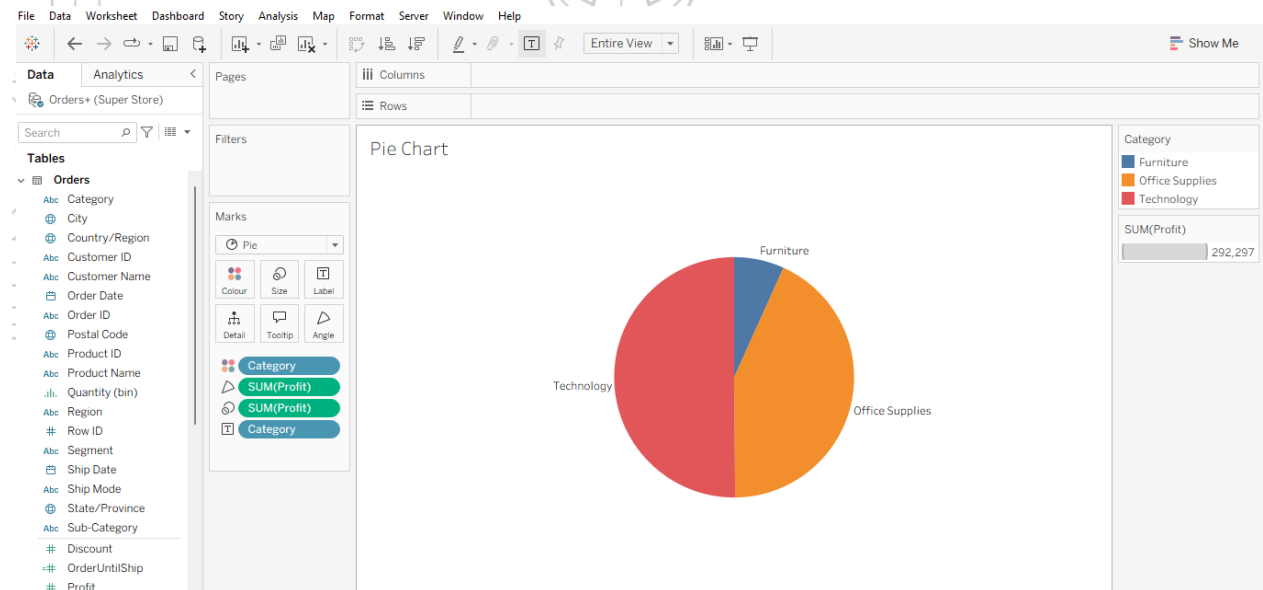
Do one of these:

- Option 1: Go to Fit in the toolbar > select Entire View
- Option 2: Click Size on the Marks card and drag the slider right

4. Add labels

- Drag Category to Label on the Marks card.
- Each slice will now show Furniture, Office Supplies, and Technology.

Out put:



Sheet 10 : Scatter Plot

1. Create the axes

- Drag Sales to Columns → this becomes the X-axis.
- Drag Profit to Rows → this becomes the Y-axis.
- You'll get a single mark scatter plot to start.

2. Color by Category

- Drag Category to Color on the Marks card.
- Each category—Furniture, Office Supplies, Technology—gets its own color.

3. Break it down by Region

- Drag Region to Detail on the Marks card.

- Now each Category splits into Central, East, South, and West.

- Total points = Categories × Regions.

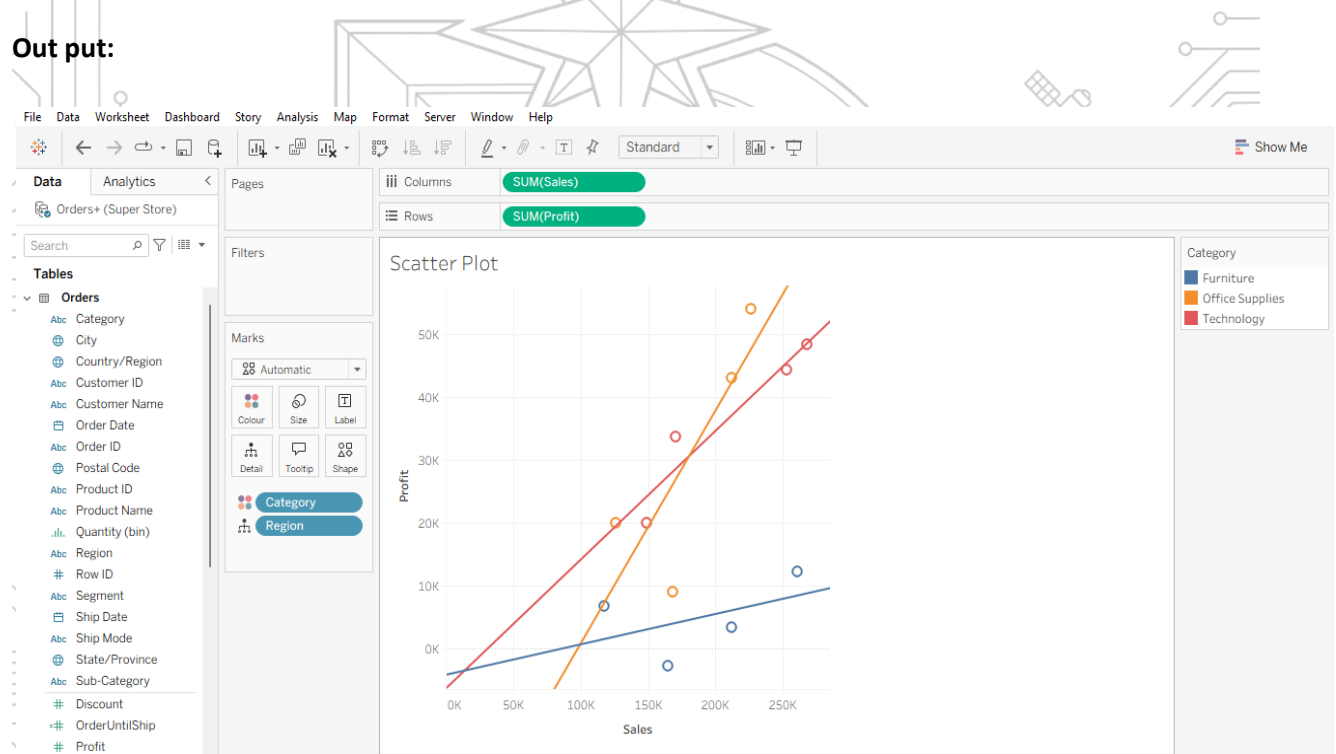
4. Add a trend line

- Open the Analytics pane on the left.

- Drag Trend Line onto the chart.

- Choose Linear to see the overall relationship between Sales and Profit.

Out put:



Sheet 11 : Text Table with Percentages

1. Build the table structure

- Drag Order Date to Columns. Tableau defaults to `YEAR(Order Date)`.

- Drag Sub-Category to Rows.

- Now you have years across the top and sub-categories down the side.

2. Show Sales values

- Drag Sales to Text on the Marks card.

- Each cell now shows `SUM(Sales)` for that year and sub-category.

3. Break it down by Region

- Drag Region to Rows and place it to the left of Sub-Category.

- Rows now look like: Central → Chairs, Central → Phones, East → Chairs, etc.

4. Convert to percentages

- Right-click SUM(Sales) on the Marks card.

- Select Add Table Calculation.

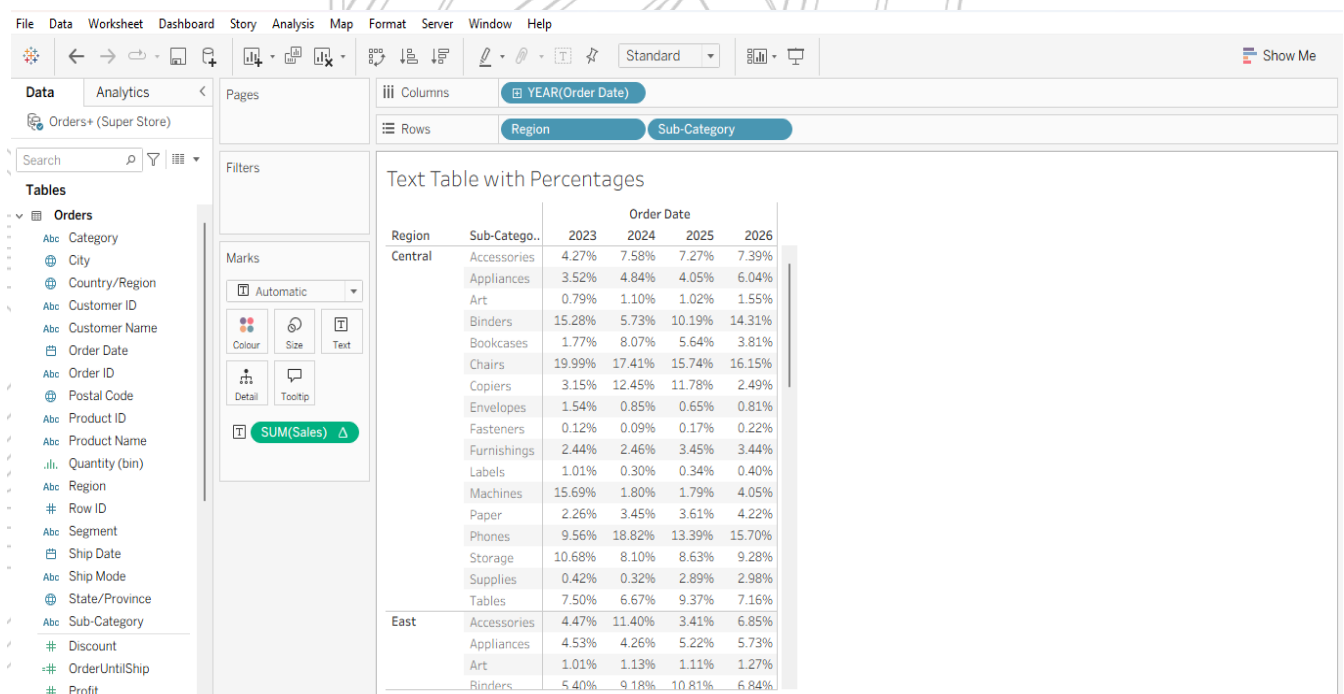
- In the dialog:

- Calculation Type: Percent of Total

- Compute Using: Pane (Down)

- Click OK.

Out put:



The screenshot shows the Tableau interface with the following configuration:

- Columns:** YEAR(Order Date)
- Rows:** Region, Sub-Category
- Marks:** SUM(Sales) (set to Percent of Total)

The resulting table is titled "Text Table with Percentages" and displays the following data:

Region	Sub-Catego..	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
	Storage	10.68%	8.10%	8.63%	9.28%
East	Supplies	0.42%	0.32%	2.89%	2.98%
	Tables	7.50%	6.67%	9.37%	7.16%
	Accessories	4.47%	11.40%	3.41%	6.85%
	Appliances	4.53%	4.26%	5.22%	5.73%
	Art	1.01%	1.13%	1.11%	1.27%
	Binders	5.40%	9.18%	10.81%	6.84%

Sheet 12 : Treemap

1. Start with Sub-Category and Sales

- Drag Sub-Category to Columns.
- Drag Sales to Rows. Tableau defaults to a bar chart.

2. Switch to Treemap

- Open Show Me in the top right.
- Select Treemap.
- Now each Sub-Category shows as a rectangle.
 - Rectangle size = Sales
 - Rectangle color = Sales by default
- Example: Large dark boxes = high sales, small light boxes = low sales.

3. Color by Ship Mode

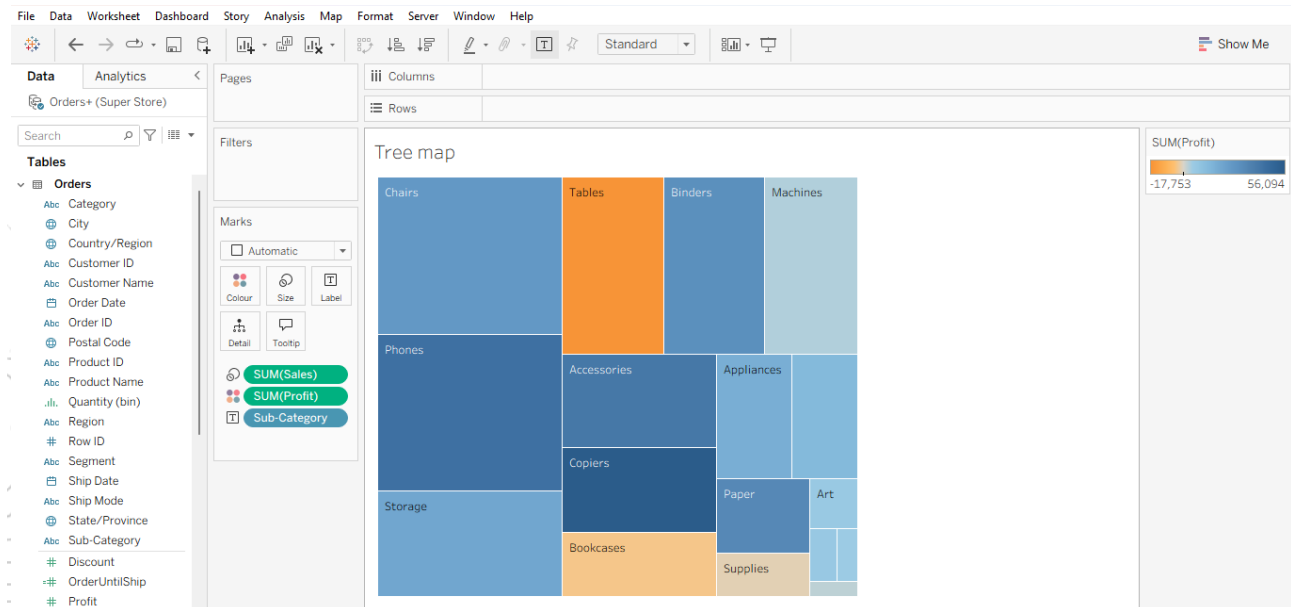
- Drag Ship Mode to Color on the Marks card.
- Rectangles now color by First Class, Second Class, Standard Class, Same Day.
- Size still reflects Sales, so you can compare sales volume and shipping method at a glance.

4. Alternative: Color by Profit

If you want profit analysis instead:

- Press Undo to remove Ship Mode from Color.
- Drag Profit to Color on the Marks card.
- Now: Size = Sales, Color = Profit. Darker/lighter colors show profit levels across sub-categories.

Out put:



Sheet 13 : Donut Chart

1. Create a helper field

- Go to Analysis > Create Calculated Field.
- Name it `Zero`, set the formula to `0`, click OK.

2. Build the base pie

- Drag `Zero` to Rows twice. You'll get two charts side by side.
- On the first one, change Marks to Pie.
- Click Fit > Entire View and increase Size on the Marks card to make it bigger.

3. Make it a dual axis donut

- Right-click the second `SUM(Zero)` on Rows > Dual Axis.
- The two pies now overlap.

4. Create the hole

- Click the second `SUM(Zero)` Marks card.
- Reduce Size slightly so it's smaller than the first pie.

- Change Color to White. This creates the donut hole.

5. Add data to the outer ring

- Click the first `SUM(Zero)` Marks card.

- Drag Region to Color so each region gets a different color.

- Drag Region and Sales to Label.

- Drag Sales to Angle so slice size matches sales.

6. Add total in the center

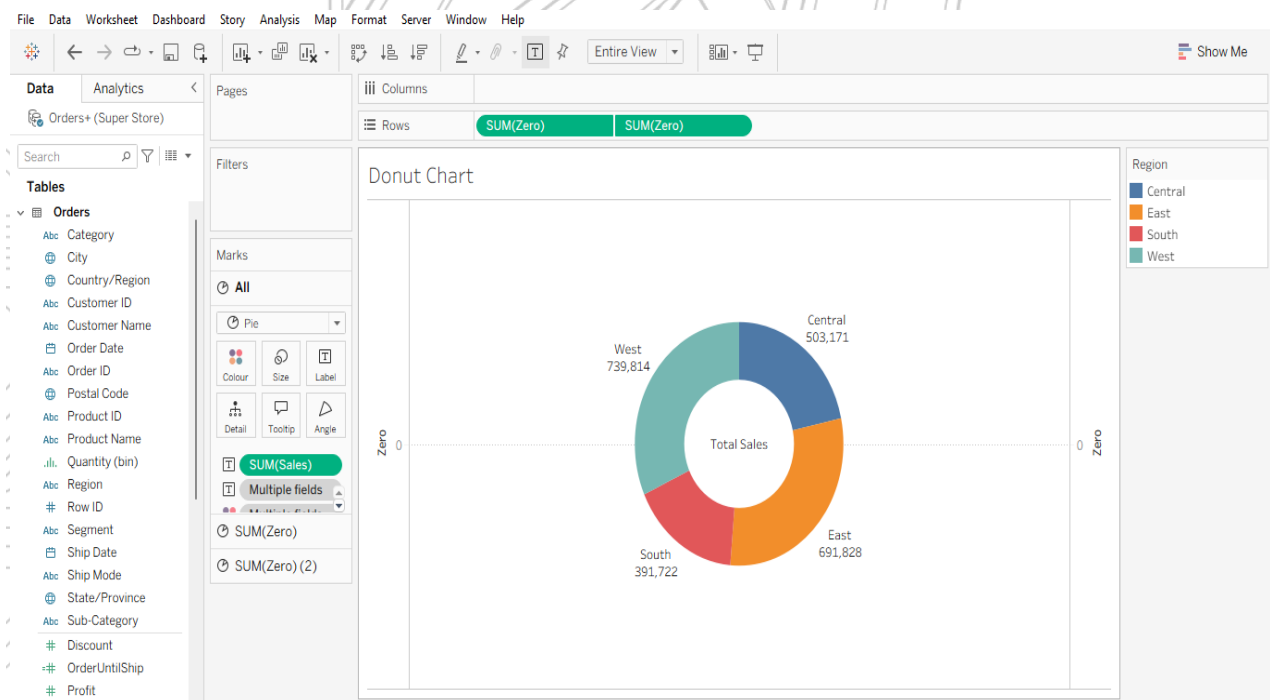
- Click the second `SUM(Zero)` Marks card.

- Drag Sales to Label.

- Click Label > Text, then the ... button.

- Type `Total Sales` above the field and click OK.

Out put:



Sheet 14 :Map Chart

1. Create the base map

- Drag State from the Data Pane onto the workspace.
- Tableau automatically plots it using Latitude and Longitude.

2. Show Sales with color

- Drag Sales to Color on the Marks card.
- States/locations now appear in different colors based on sales value.

3. Pick your map style

- On the Marks card, click Automatic.
- Choose Map for circles on locations, or Filled Map to color entire states.

4. Adjust the color palette

- Click Color > Edit Colors on the Marks card.
- Pick a palette like Purple and click OK.

5. Add labels

- Drag State and Sales to Label on the Marks card.
- Each area will now show its name and sales value.

6. Add a Region filter

- Drag Region to the Filters shelf.
- Select the regions you want to view and click OK.
- This lets you filter the map by region.

7. Show Profit on hover

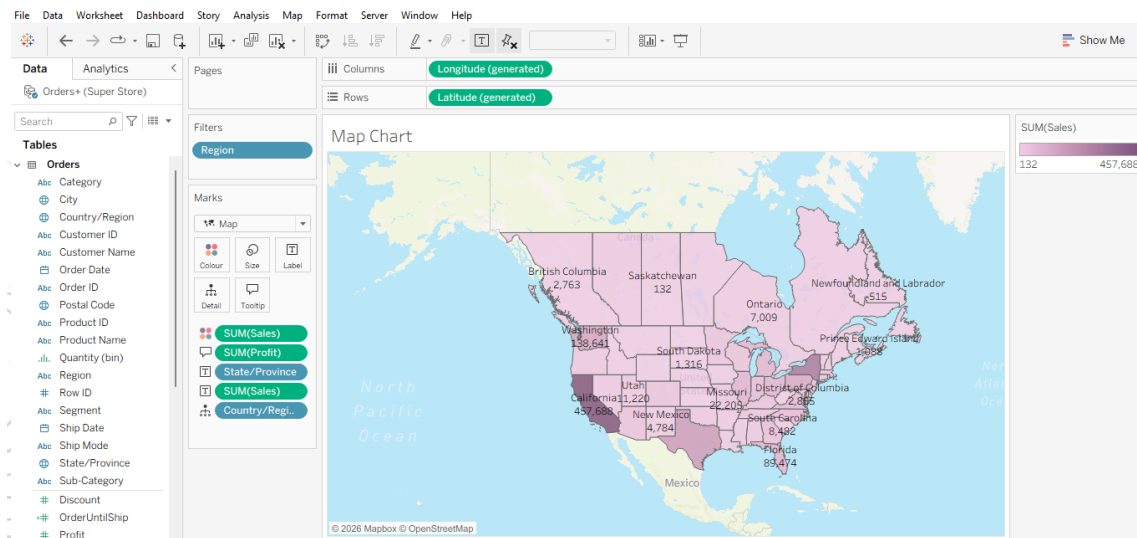
- Drag Profit* to *Tooltip.
- Now when you hover over a state, you'll see both Sales and Profit.

8. Tidy up the map look

- Go to Map > Map Layers.

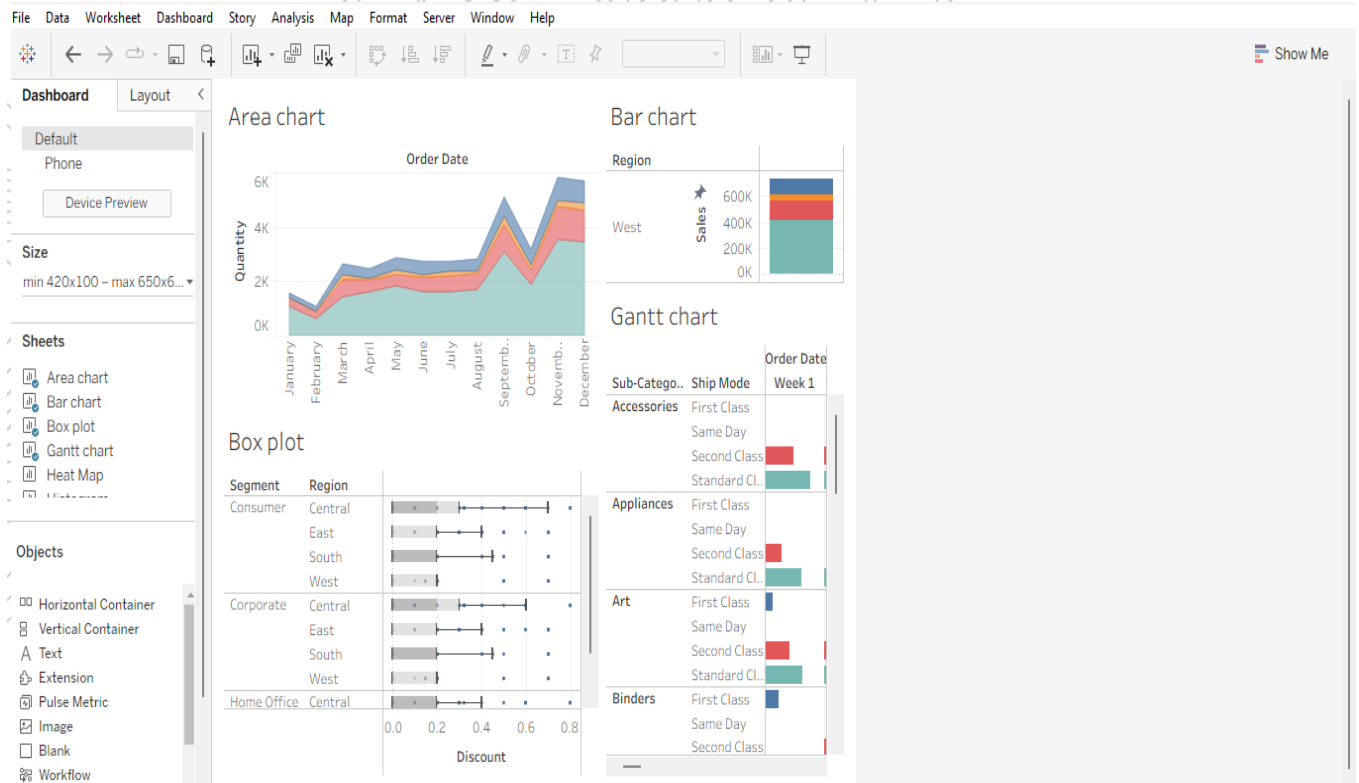
- Set Background style to Outdoors and Washout to 70% for a cleaner view.

Out put:

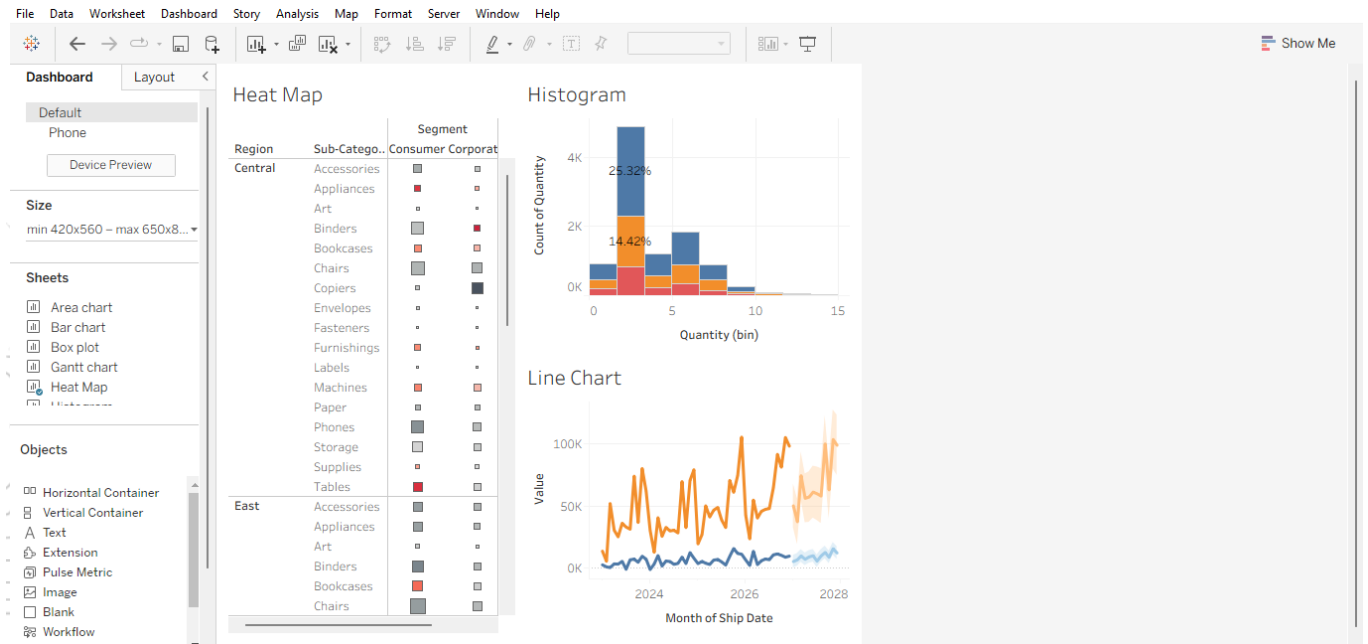


DASHBOARDS

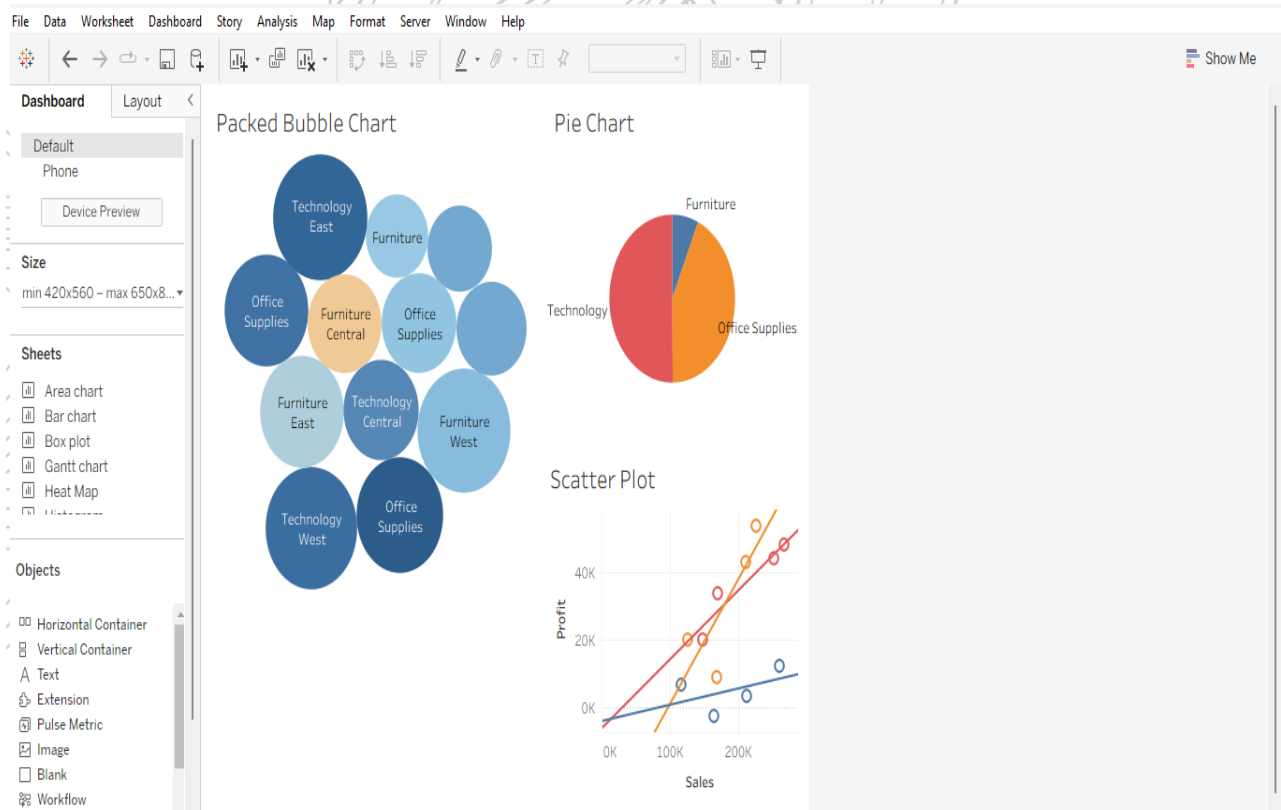
DASHBOARD-01



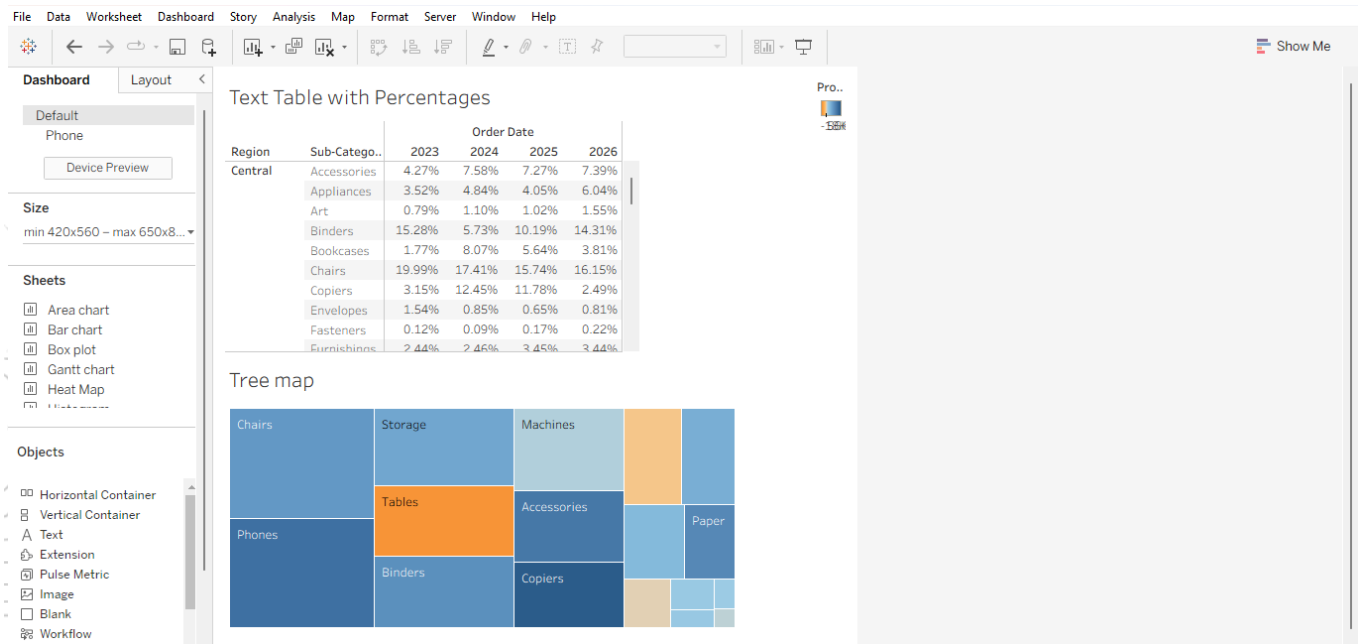
DASHBOARD-02



DASHBOARD-03



DASHBOARD-04



DASHBOARD-05

